GUIDE 6

Interaction Visualization

Goal: Visualize protein-ligand interactions using PyMOL, PLIP, and Discovery Studio on the representative frame selected in Guide 3.

Starting point: rep_structure_full_10k.pdb from Guide 3 (Step 7). This is the representative frame containing all atoms (protein, ligand, water, lipid).

PHASE 1: PyMOL — Binding Site Visualization

1.1. Load and Clean

```
load rep_structure_full_10k.pdb
hide everything
bg_color white
show cartoon, chain B
color gray80, all
```

1.2. Show Ligand and Key Residues

Show mexiletine and binding site residues as sticks. Adjust resid numbers to match your system:

```
# For P-loop binding (neutral mexiletine)
show sticks, resi 347+348+349+353+355 and chain B
show sticks, resname UNL

# For Domain III/IV binding (protonated mexiletine)
show sticks, resi 1714+1423+1424+1400+1399+1717 and chain F+H
show sticks, resname UNL
```

1.3. Color Residues by Contact Frequency

Color each residue according to its trajectory contact frequency. Adjust colors to match your results:

```
# Mexiletine
color magenta, resname UNL

# P-loop residues (neutral system)
color tv_green, resi 348 and chain B    # PRO348 88%
color tv_yellow, resi 349 and chain B   # ASP349 74.6%
color orange, resi 353 and chain B      # THR353 88.8%
color red, resi 355 and chain B         # PHE355 49.8%
color cyan, resi 347 and chain B        # LYS347 0%

# Domain IV residues (protonated system)
color red, resi 1714 and chain H        # ASP1714 67.9%
color orange, resi 1423 and chain F     # ASP1423 65.4%
color tv_green, resi 1400 and chain F   # VAL1400 54.9%
color tv_yellow, resi 1424 and chain F  # ILE1424 25.5%
```

1.4. Add Labels

```
set label_size, 16
set label_color, black
set label_bg_color, white
```

```
# Adjust resid and chain to match your system
label resi 348 and name CA and chain B, "PRO348 88%"
label resi 349 and name CA and chain B, "ASP349 74.6%"
label resi 353 and name CA and chain B, "THR353 88.8%"
label resi 355 and name CA and chain B, "PHE355 49.8%"
label resi 347 and name CA and chain B, "LYS347 0%"
```

1.5. Zoom and Save

```
zoom (resi 347 or resi 348 or resi 349 or resi 353 or resi 355 or resname
UNL), 5
ray 1200, 900
png /path/to/binding_site.png, dpi=300
```

Note: For Domain III/IV system, replace the resid list in the zoom command with the relevant residues (1714, 1423, 1424, etc.).

PHASE 2: PLIP — Interaction Analysis

PLIP (Protein-Ligand Interaction Profiler) automatically detects and classifies all interactions between the ligand and the protein in the representative frame.

2.1. Prepare Input PDB

PLIP requires a clean PDB with only protein and ligand (no water, no lipid). Extract from the representative frame:

```
grep '^ATOM\|^HETATM' rep_structure_full_10k.pdb | \
  grep -v 'TIP3\|OPC\|POPC\|SOD\|CLA\|POT' > rep_plip_input.pdb
```

2.2. Run PLIP

Via the PLIP web server (recommended):

- URL: https://plip-tool.biotec.tu-dresden.de/plip-web/plip/index
- Upload rep_plip_input.pdb
- Click Analyze
- Download results (HTML or XML)

Or via command line if PLIP is installed:

```
plip -f rep_plip_input.pdb -o plip_results/ -t
```

2.3. Interpret PLIP Output

PLIP reports the following interaction types:

- Hydrophobic interactions — distance and atom indices
- Hydrogen bonds — distance D-H···A, angle, donor/acceptor atoms, side chain vs backbone
- Salt bridges — charged group interactions
- Pi-stacking (parallel and T-shaped) — aromatic ring interactions
- Pi-cation interactions

Note: Always check whether H-bonds involve backbone or sidechain atoms. PLIP reports this in the 'Side chain' column. Backbone H-bonds with K347 do not indicate sidechain contact.

PHASE 3: Discovery Studio — 2D Interaction Diagram

Discovery Studio Visualizer (free version) generates 2D ligand interaction diagrams showing all protein-ligand contacts with interaction types and distances.

3.1. Load Structure

- Open Discovery Studio Visualizer
- File → Open → select rep_plip_input.pdb

3.2. Define Receptor and Ligand

- In the left panel: Receptor-Ligand Interactions → Define the receptor and ligand
- Receptor: select the protein model
- Ligand: Step through ligands → select UNL

3.3. Display Interactions

- Click 'Ligand Interactions' to show interactions
- Click 'Show 2D Diagram' to open the 2D interaction map
- Use '+ Expand' to show additional weaker interactions (alkyl, pi-alkyl)

3.4. Interaction Criteria (Default Cutoffs)

The following cutoffs are used by Discovery Studio by default. These can be verified via Interaction Options → Advanced:

- Alkyl / Pi-Alkyl: 5.50 Å
- H-bond (D-H···A): 3.40 Å
- Charge-charge: 5.60 Å
- Pi-Cation: 5.00 Å
- Pi-Pi: 6.00 Å

Note: 'Expand' in DS increases the number of interaction types shown (e.g., adds C-H bonds, carbon interactions), not the distance cutoff. The cutoffs above are fixed defaults.

3.5. Save the Diagram

- Right-click the 2D diagram → Save image as PNG
- For reporting: use 'expanded' view to show all interactions, and note that DS uses default cutoffs as listed above

PHASE 4: Comparing PLIP and Discovery Studio

PLIP and DS may show different interactions for the same residue. This is expected:

- PLIP H-bond with LYS347 backbone — involves the backbone oxygen (O atom), not the sidechain. This is consistent with 0% sidechain contact in trajectory analysis.
- DS van der Waals with LYS347 — detects close C-atom proximity, which can involve both backbone and sidechain carbons.
- Use trajectory contact analysis (Guide 3, Phase 4) as the ground truth for contact frequency. PLIP and DS provide single-frame structural context.

Note: Always cross-check PLIP/DS results with the trajectory contact profile. A residue appearing in the DS diagram does not necessarily mean it is a dominant contact — check the percentage from MDAnalysis.